Panskura Banamali College (Autonomous)

M.Sc. 1st Semester Examination -2022 (Online Mode)

Subject : Computer Science Paper: COS-191

Paper Name: Data Structure Lab

F.M.: 40 Time :2 Hours

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf .For Example – candidate_id: 2021PG009967, paper_code : COS-191 , then file name will be :: 2021PG009967- COS-191.pdf for question 'A' and 2021PG009967- COS-191.mp3 for question 'B']

A.Answer any one question 1×30=30

1. Write a program to implement Matrix multiplication .
2. Write a program to implement Binary search.
3. Write a program to implement a stack using two queues .
4. Write a program to implement Insertion sort.
5. Write a program to implement Merge sort.
6. Write a program to implement Insertion sort.
7. Write a program to implement a stack using Linked List .
8. Write a program to Sort an stack using another stack .

Marks distribution:

a. Problem Definition -	10 Marks
b. Algorithm/Flow Chart/ Pseudo-code–	15 Marks
c. Program listing -	03 Marks
d. Output/Result with discussion-	02 Marks

B.Answer any fivequestion 5*2=10

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What is linear data structure ?
2. What is a stack ?
3. What is d-queue?
4. What is Tower-of-Hanoi problem?
5. What is the advantages of binary search ?
6. What is hash function ?
7. What are the applications of linked list?
8. Define AVL tree.
9. Define BST.
10. What is polish notations?

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